Chemistry Study Guide

Physical Chemistry

Atomic Structure

- Rutherford's Atomic Model: Main postulates, experimental observations, conclusions, and limitations.
- Bohr's Atomic Model: Postulates and limitations/defects.
- Spectra of H-atom: Lyman, Balmer, Paschen, Brackett, Pfund.
- Derivation of De-Broglie's equation.
- Heisenberg's uncertainty principle.
- Quantum Numbers: Importance/significance of Principal, Azimuthal, Magnetic, and Spin quantum numbers.
- Difference between orbit and orbitals.
- Aufbau principle, Pauli's exclusion principle, and Hund's rule.

Periodic Table

- Modern and Mendeleev's Periodic Table: Law, advantages, and disadvantages.
- Periodicity and periodic trends: Atomic radii, ionic radii, ionization energy, electron affinity, and metallic characters.

Stoichiometry

- Numerical related to limiting reactant.
- Laws of Stoichiometry: Statement and numericals.
- VVI derivation: Mol. mass = 2 x V.D.

Chemical Bonding

- VSEPR Theory and Valence Bond Theory
- Hybridization
- Hydrogen bonding
- Vander-Waal's force

Oxidation and Reduction

- Definition of Oxidation and Reduction
- Balancing equations (Fix 5 marks)
- Electrolysis: Faraday's 1st and 2nd law statement and numericals.

States of Matter

Gas: Kinetic theory and its postulates; Gas laws (Charles' law statement/formula); Dalton's law of partial pressure; Graham's law of diffusion.

Liquid: Surface tension and viscosity.

Solid: Difference between Amorphous and Crystalline solids; Efflorescent, Deliquescent, and Hygroscopic solids; Water of crystallization.

Chemical Equilibrium

- Le Chatelier's principle and numericals
- Relation between Kp and Kc

Inorganic Chemistry (20 Marks)

Chemistry of Non-metals

Main Focus: Hydrogen, Oxygen/Ozone, Nitrogen.

1. Isotopes
2. Difference between Atomic hydrogen and Nascent hydrogen
3. Reactions related to Hydrogen and its applications
4. Heavy water
5. Oxygen: Oxides (Acidic, Basic, Amphoteric, Peroxide); Application of Oxygen and H_2O_2
6. Ozone: Resonating structure, Test, Cause/effect/control of ozone depletion
7. Nitrogen: Inertness of Nitrogen vs Active nitrogen
8. Ammonia: Properties, Test, Applications, and Harmful effects; NH_3 vs PH_3 comparison
9. Nitric Acid: Reaction with metals/non-metals; Ring test (VVI); Uses and harmful effects
10. Halogens: Bleaching actions; Comparative study ($I < Br < F < Cl$); Test of all
11. Carbon: Crystalline vs Amorphous examples; Properties/uses of Diamond, Graphite, Fullerene; Carbon monoxide
12. Hydrogen sulphide (H_2S): Preparation in Kipp's apparatus
13. Sulphuric Acid (H_2SO_4): Acidic and Oxidizing agent properties; Bleaching action of SO_2 vs Cl_2

Chemistry of Metals

1. Definitions: Gangue, Matrix, Flux, and Slag
2. Metallurgy: Froth flotation, calcination, roasting, smelting, electro-refining
3. Alkaline Metals & Alkaline Earth Metals: Molecular formula and use
4. Extraction of Sodium by Down's process
5. Use of NaOH and Na_2CO_3
6. Bleaching powder and Plaster of Paris
7. Relative strength of solubility and stability

Organic Chemistry (18 Marks)

Basic Concepts & Fundamentals

- Vital Force Theory
- Difference between organic and inorganic compounds

- Functional Group: Definition and examples
- Homologous Series: Definition and 5 characteristics
- Octane and Cetane number
- Cracking and reforming (statement)
- Nomenclature: IUPAC carbon chain
- Lassaigne's test (VVI) for N, S, and Halogens (X)
- Isomerism
- Reaction Mechanics: Homolytic/Heterolytic Fission; Electrophile/Nucleophile; Inductive Effect (+I and -I); Resonance Effect (+R and -R)

Hydrocarbons

- Preparation of Alkanes and Alkenes
- Reactions: Wurtz reaction, Decarboxylation, Catalytic hydrogenation, Dehydration, Dehydrohalogenation
- Markovnikov's rule and Anti-Markovnikov's rule
- Baeyer's test and Bromine water test
- Comparative study of Alkane, Alkene, and Alkyne

Aromatic Hydrocarbons

- Huckel's Rule
- Resonance in Benzene
- Kekule's Structure and its demerits
- Preparation / Orientation

Applied Chemistry (9 Marks)

Industrial Manufacture of:

- Ammonia by Haber's process
- Nitric Acid by Ostwald's process
- Sulphuric Acid by Contact process (Badische process)
- Sodium Hydroxide by Diaphragm cell
- Production of Urea and Fertilizers

Bio-Organic

1. Na-K pump
2. Heavy metal toxicity (MCQ)
3. Macro and Micro nutrients